

Test Form C

Name _____ Date _____

Chapter 11

Class _____ Section _____

1. A vector $\mathbf{v}$ has initial point $(-1, -3)$ and terminal point $(2, 1)$. Find the unit vector in the direction of $\mathbf{v}$.

(a) $\frac{1}{\sqrt{5}}\mathbf{i} - \frac{2}{\sqrt{5}}\mathbf{j}$

(b) $\frac{3}{5}\mathbf{i} + \frac{4}{5}\mathbf{j}$

(c) $\frac{4}{5}\mathbf{i} + \frac{3}{5}\mathbf{j}$

(d) $\frac{1}{\sqrt{5}}(\mathbf{i} + 2\mathbf{j})$

(e) None of these

2. Find a vector with magnitude $\sqrt{10}$ in the direction of $\mathbf{v} = \langle -1, 1 \rangle$.

(a) $\left\langle -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right\rangle$

(b) $\langle -\sqrt{10}, \sqrt{10} \rangle$

(c) $\langle -\sqrt{5}, \sqrt{5} \rangle$

(d) $\langle -\sqrt{2}, \sqrt{2} \rangle$

(e) None of these

3. Forces with magnitudes of 200 and 300 pounds act on a machine part at angles of 60° and -45° , respectively, with the positive x -axis. Find the direction of the resultant forces.

(a) -7.1°

(b) 7.1°

(c) 82.8°

(d) 7.5°

(e) None of these

4. Find the standard equation for the sphere that has points $(4, -3, 5)$ and $(-6, 1, -1)$ as endpoints of a diameter.

(a) $(x + 1)^2 + (y + 1)^2 + (z - 2)^2 = 38$

(b) $(x - 1)^2 + (y - 1)^2 + (z + 2)^2 = 36$

(c) $(x + 1)^2 + (y + 1)^2 + (z - 2)^2 = 36$

(d) $(x - 5)^2 + (y + 2)^2 + (z - 3)^2 = 6$

(e) None of these

5. Determine which vector is parallel to the vector $\mathbf{v} = \langle 2, -3, -1 \rangle$.

(a) $\langle 4, 6, -2 \rangle$

(b) $\left\langle -\frac{2}{3}, 1, \frac{1}{3} \right\rangle$

(c) $\left\langle 1, -\frac{3}{2}, \frac{1}{2} \right\rangle$

(d) $\langle 6, -9, 3 \rangle$

(e) None of these

6. Find the unit vector in the direction of the vector $\mathbf{v} = 2\mathbf{i} + \mathbf{j} + 2\mathbf{k}$.

(a) $\frac{2}{\sqrt{5}}\mathbf{i} + \frac{1}{\sqrt{5}}\mathbf{j} + \frac{2}{\sqrt{5}}\mathbf{k}$

(b) $\frac{2}{9}\mathbf{i} + \frac{1}{9}\mathbf{j} + \frac{2}{9}\mathbf{k}$

(c) $\mathbf{i} + \mathbf{j} + \mathbf{k}$

(d) $\frac{2}{3}\mathbf{i} + \frac{1}{3}\mathbf{j} + \frac{2}{3}\mathbf{k}$

(e) None of these

7. Find $\mathbf{u} \cdot \mathbf{v}$ if $\|\mathbf{u}\| = 40$, $\|\mathbf{v}\| = 15$, and the angle between vectors $\mathbf{u}$ and $\mathbf{v}$ is $\frac{2\pi}{3}$.
- (a) $-300\sqrt{3}$ (b) $\frac{900}{\pi}$ (c) 300
(d) -300 (e) None of these
8. Find the vector component of $\mathbf{u}$ orthogonal to $\mathbf{v}$ for $\mathbf{u} = \mathbf{i} + 2\mathbf{j}$ and $\mathbf{v} = -4\mathbf{i} + 4\mathbf{j}$.
- (a) $-\frac{1}{2}\mathbf{i} + \frac{1}{2}\mathbf{j}$ (b) $\frac{3}{2}\mathbf{i} + \frac{3}{2}\mathbf{j}$ (c) $-\frac{1}{\sqrt{2}}\mathbf{i} + \frac{1}{\sqrt{2}}\mathbf{j}$
(d) $\frac{1}{2}\mathbf{i} + \frac{5}{2}\mathbf{j}$ (e) None of these
9. Calculate the angle that vector $\mathbf{v} = 3\mathbf{i} - 5\mathbf{j} + \mathbf{k}$ makes with the positive y -axis.
- (a) 59.5° (b) 32.3° (c) 147.7°
(d) 80.3° (e) None of these
10. $\mathbf{v} \cdot \mathbf{v} =$ _____
- (a) $\|\mathbf{v}\|$ (b) $\frac{\mathbf{v}}{\|\mathbf{v}\|}$ (c) $\|\mathbf{v}\|^2$
(d) $\text{proj}_{\mathbf{u}}\mathbf{v}$ (e) None of these
11. Calculate $\mathbf{u} \times \mathbf{v}$ for $\mathbf{u} = 3\mathbf{i} + 4\mathbf{j} - \mathbf{k}$ and $\mathbf{v} = 2\mathbf{i} - \mathbf{j} + 3\mathbf{k}$.
- (a) $-11\mathbf{i} + 11\mathbf{j} + 11\mathbf{k}$ (b) $13\mathbf{i} - 7\mathbf{j} + 5\mathbf{k}$ (c) $11\mathbf{i} + 11\mathbf{j} - 11\mathbf{k}$
(d) $11\mathbf{i} - 11\mathbf{j} - 11\mathbf{k}$ (e) None of these
12. Find the area of the parallelogram having vectors $\mathbf{v}_1 = -\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$ and $\mathbf{v}_2 = 3\mathbf{i} - 2\mathbf{j} + \mathbf{k}$ as adjacent sides.
- (a) 3 (b) 10 (c) $\sqrt{101}$
(d) $\sqrt{69}$ (e) None of these
13. Find an equation of the plane determined by the points $(1, 2, -3)$, $(2, 3, 1)$, and $(0, -2, -1)$.
- (a) $18\mathbf{i} - 6\mathbf{j} - 3\mathbf{k} = 0$ (b) $6x - 2y - z = 0$ (c) $6x - 2y - z = 5$
(d) $14x + 6y + 3z = 17$ (e) None of these
14. Find the point of intersection of the line given by the parametric equations $x = 3 + 2t$, $y = 7 + 8t$, and $z = -2 + t$ with the yz -plane.
- (a) $(0, -5, -\frac{7}{2})$ (b) $(0, \frac{37}{3}, -\frac{4}{3})$ (c) $(3, 7, -2)$
(d) $(-7, -33, -7)$ (e) None of these

15. Find a generating curve for the surface of revolution given by $2x^2 + y^2 + z^2 = 3x$.
- (a) $y = \sqrt{3x - 2x^2}$ (b) $z = \sqrt{3x - 2x^2}$ (c) $x = y^2 + z^2$
(d) Both (a) and (b) (e) None of these
16. Write the equation in standard form and identify the quadric surface given by $4x^2 - 9y^2 - 36z = 0$.
- (a) $36z = 4x^2 - 9y^2$, Cone
(b) $z = \frac{x^2}{9} - \frac{y^2}{4}$, Hyperbolic paraboloid
(c) $z = \frac{x^2}{9} - \frac{y^2}{4}$, Hyperboloid of one sheet
(d) $x^2 = \frac{9}{4}y^2 + 9z$, Hyperbolic paraboloid
(e) None of these
17. Express the spherical coordinate point $\left(6, -\frac{\pi}{6}, \frac{\pi}{3}\right)$ in cylindrical coordinates.
- (a) $\left(\frac{3}{2}, -\frac{3\sqrt{3}}{2}, 3\right)$ (b) $\left(-\frac{3\sqrt{3}}{2}, \frac{3}{2}, 3\right)$ (c) $\left(3\sqrt{3}, 3, \frac{\pi}{3}\right)$
(d) $\left(3\sqrt{3}, -\frac{\pi}{6}, 3\right)$ (e) None of these
18. Find a rectangular equation for the graph represented by the spherical equation $\rho = 2 \cos \phi$.
- (a) $x^2 + y^2 + z^2 = 2x$ (b) $x^2 + y^2 + z^2 = 2z$ (c) $y^2 + z^2 - x^2 = 0$
(d) $z = 2x^2$ (e) None of these